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Studierichting: Informatica
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$$\int \ln|x^2+2|dx$$

$$u = \ln|x^2+2| \Rightarrow Du = \frac{2x}{x^2+2}$$

$$Dv = dx \Rightarrow v = xdx$$

$$\Rightarrow x\ln|x^2+2| - \int \frac{2x^2}{x^2+2}dx$$
$$= x\ln|x^2+2| - \int \frac{2(x^2+2)-4}{x^2+2}dx$$
$$= x\ln|x^2+2| - 2x + 2\sqrt{2}\,\mathrm{Bgtan}\left(\frac{x}{\sqrt{2}}\right) + C$$

References